

Nome: Marcos Vinícius da Silva

Matrícula: 16-95709

$$1 \rightarrow Z_1 = \frac{45 - 50}{10} = \frac{-5}{10} = -0,5$$

$$Z_2 = \frac{62 - 50}{10} = \frac{12}{10} = 1,2$$

$$P(45 < X < 62) = P(-0,5 < Z < 1,2) = P(Z < 1,2) - P(Z < -0,5) = 0,8849 - 0,3085 =$$

$$\boxed{0,5764}$$

$$2 \rightarrow P(Z > K) = 0,3025$$

$$\textcircled{a} \quad 0,3025 = 1 - (P(Z < K))$$

$$P(Z < K) = 1 - 0,3025$$

$$P(Z < K) = 0,6975$$

$$\boxed{K = 0,52}$$

$$\textcircled{b} \quad P(K < Z < -0,18) = 0,4199$$

$$1 - 0,5924 = 0,4076$$

$$0,4076 - 0,4199 = 0,0089$$

$$1 - 0,0089 = 0,9911$$

$$\boxed{K = -2,39}$$

$$3 \rightarrow Z = \frac{362 - 300}{50} = 1,24$$

$$P(Z > 1,24) = 1 - 0,8925 = \boxed{0,1075}$$

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$$4 \rightarrow 0,45 = P(Z < -0,13)$$

$$a) X = 6(-0,13) + 40$$

$$X = 39,22$$

$$b) 0,14 = P(Z < 1,08)$$

$$X = 6(1,08) + 40$$

$$X = 46,48$$

$$5 \rightarrow Z = \frac{2,3 - 3}{0,5} = -1,4$$

$$P(X \leq 2,3) = P(Z \leq -1,4)$$

$$P(Z \leq -1,4) = 0,0808$$

$$6 \rightarrow Z_1 = \frac{778 - 800}{40} = -0,55$$

$$Z_2 = \frac{834 - 800}{40} = 0,85$$

$$F(Z_1) = 0,2912$$

$$F(Z_2) = 0,8023$$

$$P(778 < X < 834) = 0,8023 - 0,2912 = 0,5111$$

$$8 \rightarrow Z = \frac{43 - 40}{2} = 1,5$$

$$F(Z) = 0,9332$$

$$P(X > 43) = 1 - 0,9332 = 0,0668$$

